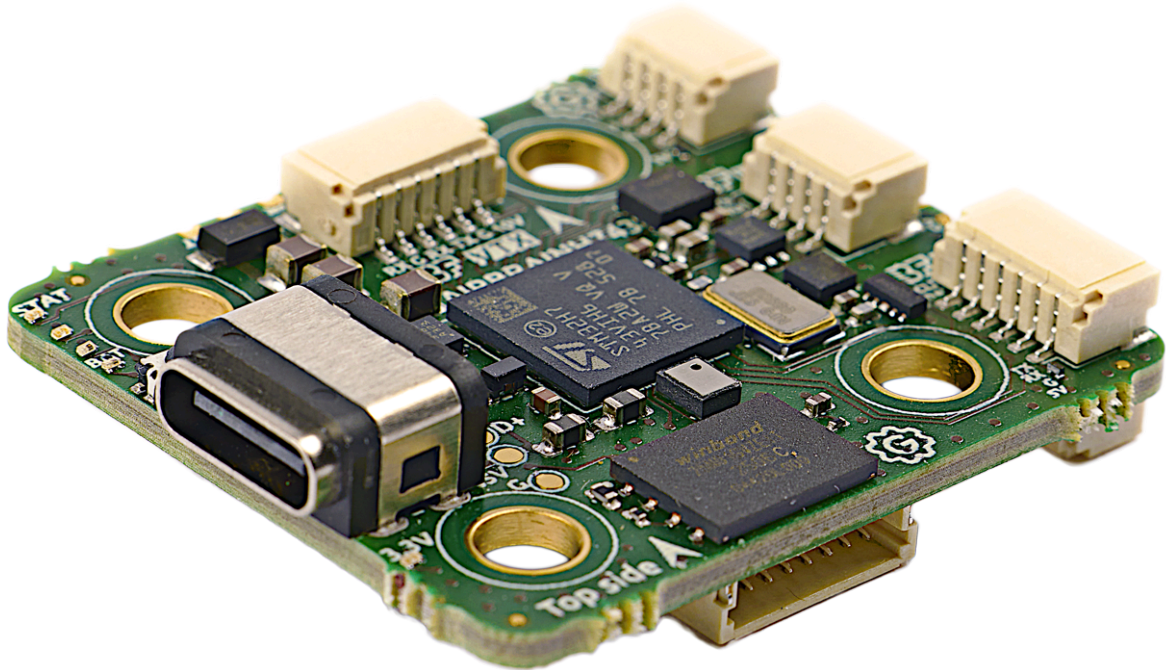




| EU-BUILT FLIGHT CONTROLLER

AirBrain H743.

Any firmware. Every mission.

Compact 20×20 flight controller built around the STM32H7. Synchronised with an ICM-42688-P IMU for low-latency, low-noise flight on Betaflight, ArduPilot, and PX4.

NDAA compliant

Made in Belgium

STM32H7

ICM-42688-P

20 × 20 mm

3-10S

AT A GLANCE — Differentiators

FIRMWARE LANES 3 stacks Upstream target in Betaflight, ArduPilot, and PX4. No vendor lock-in.	NDAA COMPLIANT 100% EU sourced Belgian assembly, trusted EU supply chain.	CLOCK-SYNCD IMU 0 drift MCU clock drives the ICM-42688-P. No software interpolation between sample and PID.	WIDE INPUT RANGE 3-10S 10- 42 V Same board across compact race builds and heavy-lift cinema rigs.
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HIGHLIGHTS — Key features

- **STM32H7 MCU.** 480 MHz Cortex-M7 with double-precision FPU, 2 MB program flash, 1 MB RAM.
- **3-axis magnetometer.** First high-performance magnetometer to ship in an ArduPilot target.
- **128 MB onboard flash.** Continuous-read NAND for blackbox / persistent log storage.
- **USB-C 2.0 receptacle.** Sealed connector body, IP67 when panel-mounted with gasket.
- **Wide BEC outputs.** 5 V @ 2 A and 10 V @ 2.5 A for GPS, RX, VTX, LED strip, cameras.
- **ICM-42688-P IMU.** Industrial-grade gyro, clock-synced with MCU, isolated power rail.
- **Barometric sensor.** Robust pressure + temperature for harsh environments.
- **EMC + ESD protected.** Pre-compliance EMC sweeps before each rev.
- **20×20 mm mount.** 36 × 31 × 10 mm overall, ≈ 8 g, low-profile stack-up.
- **NDAA compliant.** Exclusively EU / US / JP / TW components. Made in Belgium .



— How the choices shape flight behaviour

ULTRA-FAST STM32H7 MCU

High-performance 32-bit Arm® Cortex®-M7 core running at 480 MHz with double-precision floating-point unit. Drives ultra-low-latency control loops and unlocks flight performance unattainable on F4 / F7 generation autopilots. 2 MB on-chip Flash with read-while-write and 1 MB of system RAM give modern autopilot stacks (PX4, ArduPilot) generous headroom for state estimators, EKFs, and onboard logging.

HIGH-PRECISION GYROSCOPE WITH MCU CLOCK-SYNC AND ISOLATED POWER

The TDK ICM-42688-P is a 6-axis MEMS MotionTracking device combining a 3-axis gyroscope and a 3-axis accelerometer. By using the H7 MCU as the sample-clock generator for the IMU we get extremely accurate, deterministic gyroscope sample timing and very low noise, a feature only available on the H7 series of MCUs. The IMU additionally uses its own dedicated voltage regulator, isolated from the rest of the system, so commutation transients on the motor rails never reach the gyro path. Quiet logs, no random twitches at high throttle.

HIGH-PERFORMANCE 3-AXIS MAGNETOMETER

STMicroelectronics LIS2MDL, high-accuracy tilt-compensated compass for map rotation and orientation. First high-performance magnetometer to ship in an upstream ArduPilot target on this class of hardware.

HIGH-PRECISION BAROMETRIC PRESSURE SENSOR

Infineon DPS368XTSA1, robust pressure and temperature for harsh-environment altitude hold, rate-of-climb estimation, and EKF height fusion.

LARGE FLASH FOR PROGRAM AND PERSISTENT LOGGING

2 MB of on-chip Flash hosts the firmware itself with read-while-write support. Separately, a Winbond W25N01GVZEIT 128 MB NAND with continuous-read mode is fitted for blackbox logging, enough capacity to capture an entire commercial inspection mission at high log-rate without rotation.

WIDE-RANGE POWER INPUT + DUAL BEC OUTPUTS

3 – 10S LiPo support spans the full range from compact race quads up to heavy-lift cinema rigs, all on the same hardware. A TVS protection diode and reverse-polarity protection are fitted on the battery input. Two large DC/DC converters supply 5 V at 2 A and 10 V at 2.5 A, more than enough headroom for GPS, receiver, LED strip, video transmitter, cameras, and other accessories without an external regulator.

HIGH-QUALITY ACTIVE AND PASSIVE COMPONENTS

Switching-IC power topology instead of dissipative diodes, ultra-low-noise voltage regulators, and Würth inductors. Industrial-grade BOM throughout.

BATTERY MONITORING

Built-in battery voltage monitoring and dedicated current input from the ESC for closed-loop power awareness in the autopilot.

SWD DEBUG ON CONNECTOR + SMALL FORM FACTOR

Symmetric SWD debug pads exposed on a connector for in-field flashing or live debugging without disassembly. The 20 × 20 mm hole pattern with low overall profile (10 mm) drops the AirBrain into compact stack-ups without forcing redesign.

EMC + ESD PROTECTION · NDAA · MADE IN EUROPE

Every external connector is ESD-protected. Pre-compliance EMC sweeps are run before each revision so the controller survives realistic ESC / VTX / RX coexistence, not just bench-perfect conditions. The component bill of materials sources exclusively from EU / US / JP / TW countries (NDAA-compliant). Final assembly and test are done in Belgium .



— Hardware specifications, full list

Same content as the spec table, in compact bullet form for quick scan.

- 7 × UART
- 1 × I²C
- 1 × SWD debug
- 3 × programmable LED
- 1 × AUX GPIO output
- BOOT button
- BEC 5 V @ 2 A, for FC peripherals, RX, GPS, optical-flow sensors
- Betaflight standardised connectors
- 20 × 20 mm hole pattern, low profile
- Made in Belgium 
- 8 × PWM (DSHOT supported)
- 1 × SPI
- 2 × ADC (VBat, current)
- 2 × power LED (H7 power, battery present)
- 1 × AUX GPIO through transistor (buzzer –)
- USB-C 2.0 receptacle, sealed body (IP67 panel-mounted with gasket)
- BEC 10 V @ 2.5 A, for video transmitter and cameras
- Castellated pins for ESC and UART2 in addition to connectors
- NDAA compliant. EU / US / JP / TW components



— The decisions that show up in flight logs

Clock-synced IMU

The MCU clock drives the ICM-42688-P sample timing. No drift between sensor read and control loop. PID stays stable at the high gains race cine FPV pilots actually want. This feature is only available on the H7 series of MCUs.

Isolated power rails

Sensor rails are physically separated from motor rails. ESC commutation noise stays out of the gyro path. Quiet logs, no random twitches at high throttle. The IMU has its own ultra-low-noise voltage regulator.

Industrial-grade BOM

Sourced through trusted EU channels. Switch ICs instead of diodes, ultra-low-noise voltage regulators, Würth inductors. NDAA compliant.

EMC tested layout

Pre-compliance EMC sweeps before each rev. Survives realistic ESC / VTX / RX coexistence, not just bench-perfect conditions. ESD protection on every external connector.



— One controller, six platforms

RACE CINE FPV

Cinematic FPV with predictable rates and ultra-low latency on Betaflight.

LONG-RANGE MAPPING

ArduPilot autonomy with reliable GPS + magnetometer fusion.

INDUSTRIAL INSPECTION

Survey-grade flights with PX4 autopilot.

RESEARCH LABS

Hardware target upstream in PX4 and ArduPilot, no private fork needed.

FIXED-WING LONG-RANGE

ArduPilot Plane on the same hardware target.

CUSTOM AIRFRAMES

Standard upstream firmware target across stacks; integrate without forking the firmware.

**MCU & SENSORS**

MCU	STM32H743VIH6, 480 MHz, 2 MB flash, 1 MB RAM
IMU	TDK ICM-42688-P, clock-synced with MCU
Magnetometer	STMicro LIS2MDL, 3-axis
Barometer	Infineon DPS368XTSA1
Onboard storage	Winbond W25N01GVZEIT, 128 MB NAND, blackbox

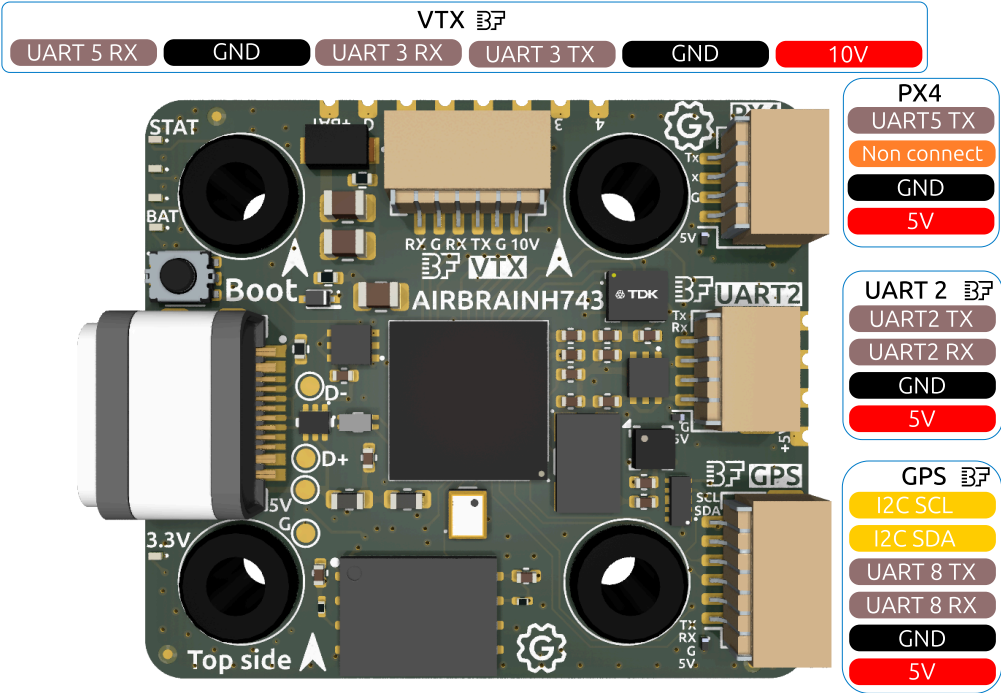
I/O

UARTs	7
Motor outputs	8 × DShot PWM
Other busses	1 × I ² C, 1 × SPI, LED-strip on connector
USB	USB-C 2.0 receptacle, sealed body (IP67 when panel-mounted with gasket)
SWD	Debug port on connector
ADC	2 × (VBat + Current)
GPIO	1 × AUX, 1 × buzzer (open-drain transistor)
Connectors	Betaflight standardised + castellated pins for ESC/UART2

POWER	
Battery input	3–10S LiPo, 10–42 V
BEC outputs	5 V @ 2 A, 10 V @ 2.5 A
3.3 V output	0.5 A max
Monitoring	Battery voltage + current sensing
Protection	TVS diode, reverse polarity, EMC, ESD

MECHANICAL	
Mounting	20 × 20 mm pattern, M3 holes
Dimensions	36 (L) × 31 (W) × 10 (H) mm
Weight	≈ 8 g
Operating temp	0 – 60 °C
Humidity	0 – 90% RH, non-condensing
Compliance	NDAA, RoHS, CE, made in Belgium

 HARDWARE REFERENCE



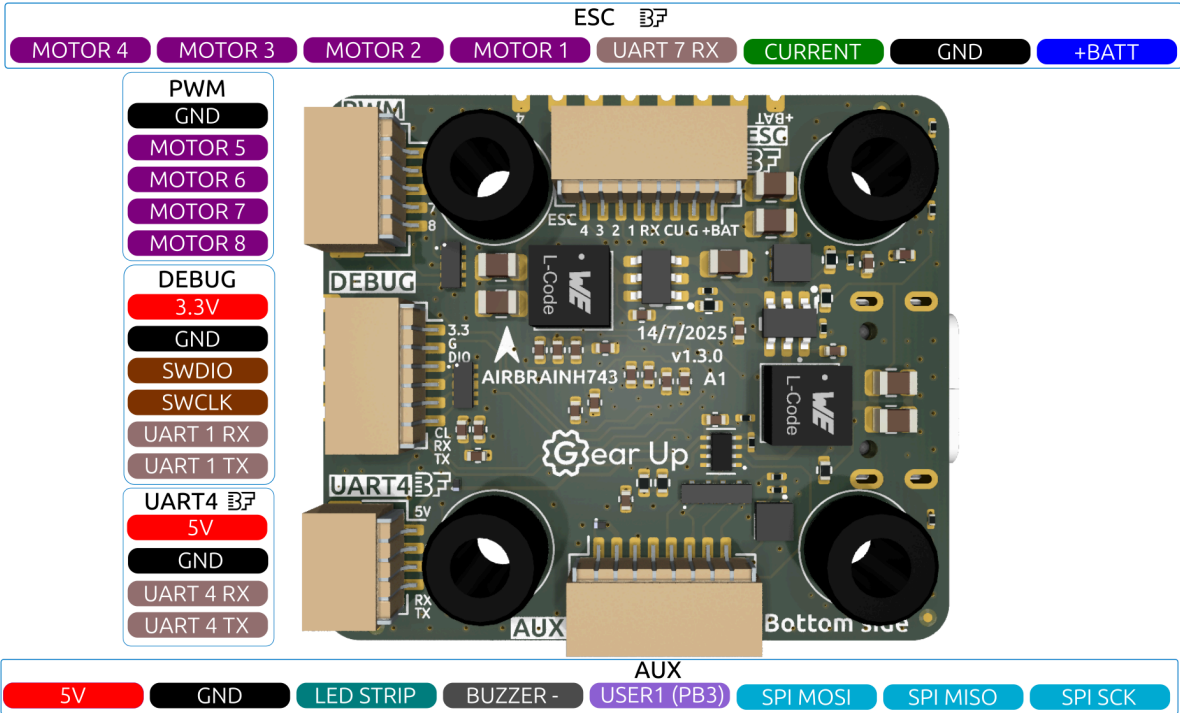
Top-side connector and pad layout. Refer to silkscreen for definitive labels.

Legend	
+BATT	
POWER	
GND	
CURRENT	
UART	
MOTOR	
LED STRIP	
BUZZER -	
AUX GPIO	
SPI	
I2C	
SBUS	
SWD	

Pin function colour legend (applies to both pinout pages).

HARDWARE REFERENCE

Pinout, bottom side



Bottom-side castellated pads + auxiliary connector.

Legend	
+BATT	
POWER	
GND	
CURRENT	
UART	
MOTOR	
LED STRIP	
BUZZER -	
AUX GPIO	
SPI	
I2C	
SBUS	
SWD	

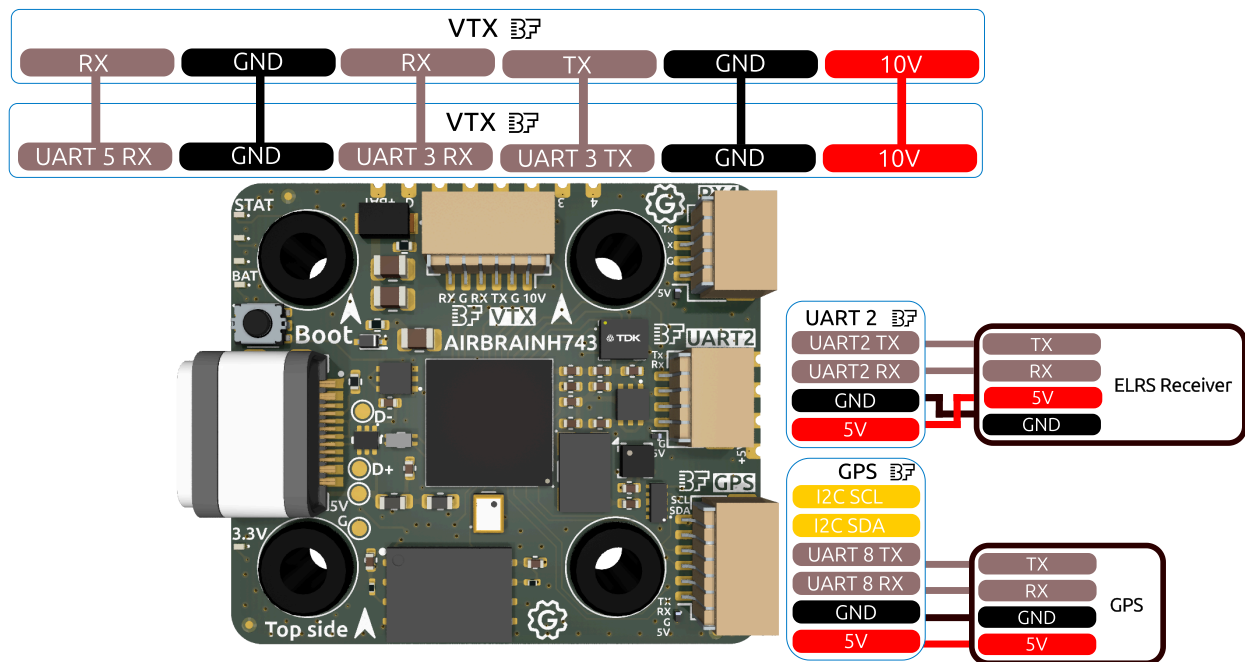
Pin function colour legend (applies to both pinout pages).



HARDWARE REFERENCE

— Connection diagram, top side

Suggested wiring for accessories (VTX, GPS, UART2, PX4 telemetry, I²C peripherals). Each block shows the connector face from the AirBrain side; route signal pins to the matching wire on the accessory.

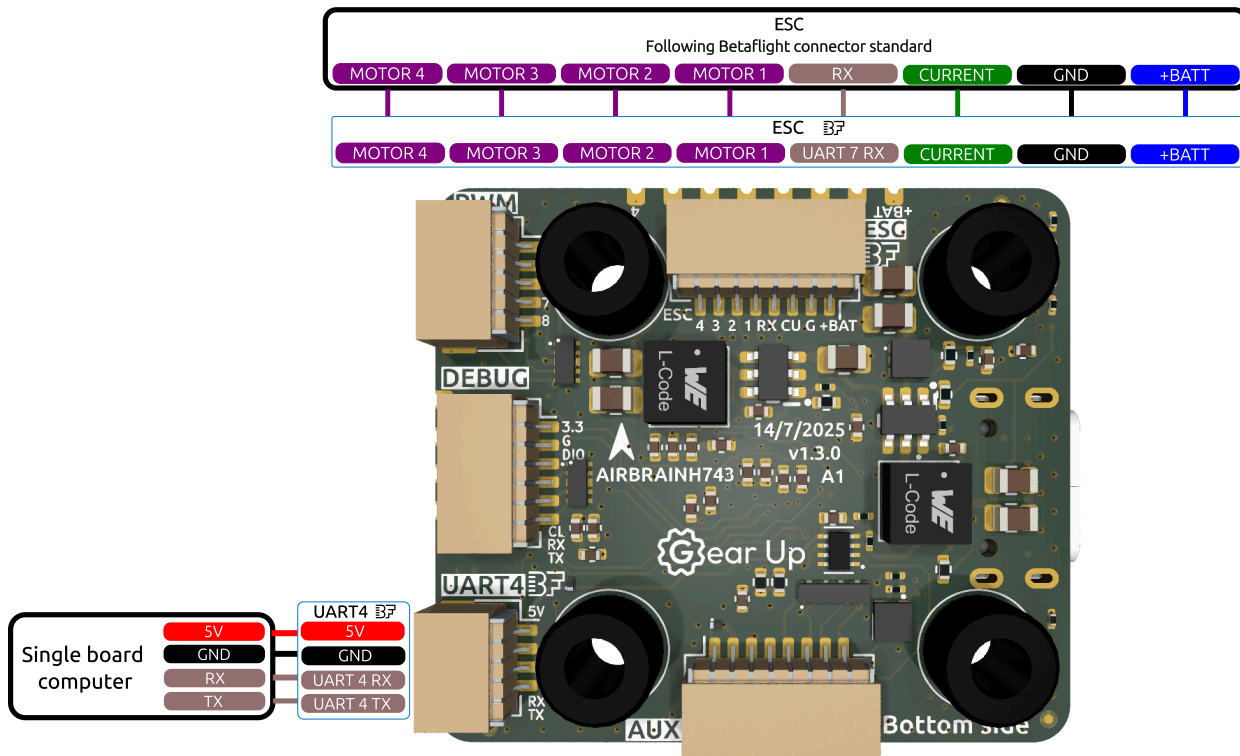


Top-side accessory wiring: VTX, GPS, UART2, PX4 telemetry, I²C.



— Connection diagram, bottom side

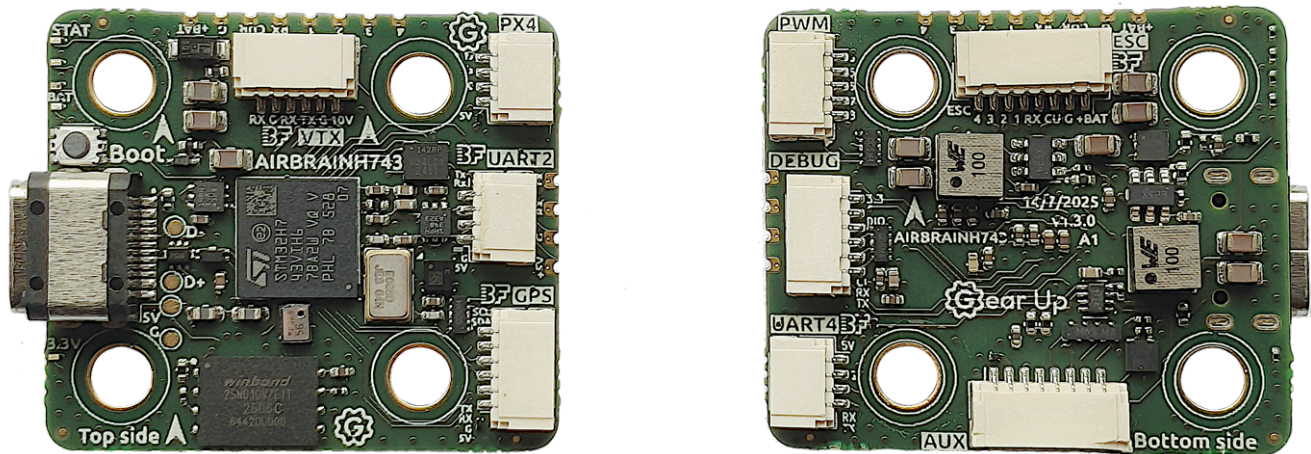
Bottom-side accessory wiring: ESC connector (Betaflight standardised), single-board-computer UART, AUX peripherals (LED strip, buzzer, SPI, I²C).



Bottom-side accessory wiring: ESC, single-board computer, AUX devices.

PRODUCT VIEWS — Front and back

Top and bottom of the AirBrain H743. Use these for reference when planning component placement, connector access, and stack-up clearance on the airframe.

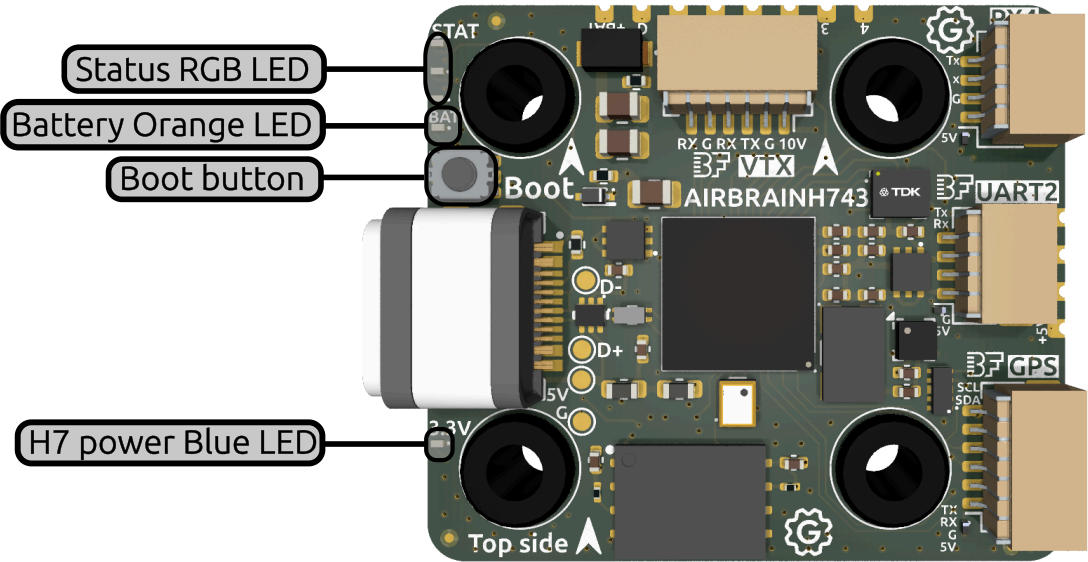


Front (left) and back (right).

MECHANICAL — Dimensions

Compact 20 × 20 mm mounting pattern, low overall height. The Pinout pages above show connector positions on the silkscreen.

Length	36 mm
Width	31 mm
Height (overall)	10 mm including connectors
Mounting pattern	20 × 20 mm, M3 holes
Weight	≈ 8 g



Top-side render with LED and button positions annotated.

H7 power	Blue	Lit whenever the STM32H7 3.3 V rail is stable. Useful as a hardware sanity check during bring-up. Off means the MCU has no power.
Battery present	Orange	Lit whenever a flight battery is connected and VBat is above the detect threshold. Independent of USB power, so you can confirm the battery harness is live without firmware running.
RGB	R / G / B	Programmable status LED. Driven by the firmware. See per-firmware patterns below.
BOOT button	.	Hold during USB connect to enter DFU mode for manual firmware flashing.

RGB LED, PER-FIRMWARE BEHAVIOUR

BETAFLIGHT

Patterns are not centrally documented; they come from the LED Strip STATUS profile defaults plus the firmware heartbeat. Customise per LED role in BF Configurator → LED Strip tab.

Armed	Solid colour for the active flight-mode profile
Disarmed	Slow heartbeat blink
Failsafe	Red strobe
Low battery	Amber pulse at the configured threshold
Beacon	High-rate strobe to locate a lost airframe

ARDUPILOT

Driven by the Notify driver. Enable in NTF_LED_TYPES; patterns follow the AP_Notify state machine. Reference: ardupilot.org/copter/docs/common-leds-pixhawk.html

Initialising	Red double-flash
Pre-arm fail	Yellow flashing
Armed (no GPS)	Blue solid
Armed (GPS lock)	Green solid
Auto / Guided	Cyan
Failsafe	Magenta strobe
Low battery	Red strobe

PX4

Driven by the RGBLED driver, auto-selected for the airframe. Patterns map directly to the PX4 vehicle state machine. Reference: docs.px4.io/main/en/getting_started/led_meanings.html

Initialising	Red, fast blink
Calibrating	White, blink
Ready to arm	Green, slow blink
Armed	Green, solid
Pre-arm fail	Yellow
Failsafe	Red, strobe / solid
Low battery	Amber, blink

Patterns are firmware defaults. Each stack lets you remap or recolour the RGB LED through its configurator (BF Configurator LED Strip tab, ArduPilot NTF parameters, PX4 RGBLED driver).

AirBrain H743 is upstream-supported in three flight stacks. The hardware target is **AIRBRAINH743**. Always flash from the official upstream channel for your stack (links per section below). The hardware definition is upstream in all three projects.

Betaflight **AIRBRAINH743**

For racing, freestyle, and race cine FPV platforms. Betaflight Configurator detects the AIRBRAINH743 target and builds the correct firmware image automatically.

- **Project** <https://betaflight.com>
- **Web configurator** <https://app.betaflight.com>
- **Desktop configurator** <https://github.com/betaflight/betaflight-configurator/releases>
- **Board documentation** <https://www.betaflight.com/docs/wiki/boards/current/AIRBRAINH743>
- **Hardware definition** <https://github.com/betaflight/config/tree/master/configs/AIRBRAINH743>

ArduPilot **AIRBRAINH743**

For autonomous, mapping, and long-range platforms. Copter, Plane, Rover, Sub, and Blimp targets all flash this hardware. Mission Planner and QGroundControl auto-detect the board over USB.

- **Project** <https://ardupilot.org>
 - **Mission Planner** <https://ardupilot.org/planner>
 - **QGroundControl** <http://qgroundcontrol.com>
 - **Latest Copter binary** <https://firmware.ardupilot.org/Copter/latest/AIRBRAINH743>
 - **Hardware definition** https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/AIRBRAINH743/README.md
-

PX4

gearup_airbrainh743

For research, advanced autonomy, and custom airframes, the AirBrain H743 is documented as *gearup_airbrainh743* in the PX4 hardware tree. Build from source via the standard PX4 toolchain, or use QGroundControl to flash a pre-built image.

- **Project** <https://px4.io>
- **Board documentation** https://docs.px4.io/main/en/flight_controller/gearup_airbrainh743
- **QGroundControl** <http://qgroundcontrol.com>
- **Hardware definition (PX4 main)** <https://github.com/PX4/PX4-Autopilot/tree/main/boards/gearup/airbrainh743>
- **Source repository** <https://github.com/PX4/PX4-Autopilot>
- **Building from source** https://docs.px4.io/main/en/dev_setup/building_px4.html
- **Latest CI binary** https://px4-travis.s3.amazonaws.com/Firmware/master/gearup_airbrainh743_default.px4
- **Upgrading from earlier build** <https://github.com/GearUp-Company/AirBrainH743/blob/main/firmware/PX4/UPGRADING.md>

Hardware repository (Gear Up)

Firmware, datasheet, mechanical adapter STEP files, and reference builds for the AirBrain H743 live in our public GitHub repository. Hardware design (schematics, BOM, layout) is proprietary to GearUp Company.

- **Repository** <https://github.com/GearUp-Company/AirBrainH743>
 - **Datasheet (PDF)** https://github.com/GearUp-Company/AirBrainH743/blob/main/datasheet/Datasheet_AirBrain.pdf
 - **Firmware mirror** <https://github.com/GearUp-Company/AirBrainH743/tree/main/firmware>
 - **Build instructions** <https://github.com/GearUp-Company/AirBrainH743/blob/main/BUILD.md>
 - **License** <https://github.com/GearUp-Company/AirBrainH743/blob/main/LICENSE.md>
 - **Issue tracker** <https://github.com/GearUp-Company/AirBrainH743/issues>
-



— Entering DFU mode for manual flashing

1. Power off the flight controller.
2. Press and hold the **BOOT** button.
3. Plug in the USB-C cable while still holding BOOT.
4. The blue Status LED will dim to half-brightness and the host OS will mount the device as a USB mass-storage / DFU device.
5. Use Betaflight Configurator, ArduPilot Mission Planner, QGroundControl, or `dfu-util` directly to flash.



OPERATING ENVELOPE

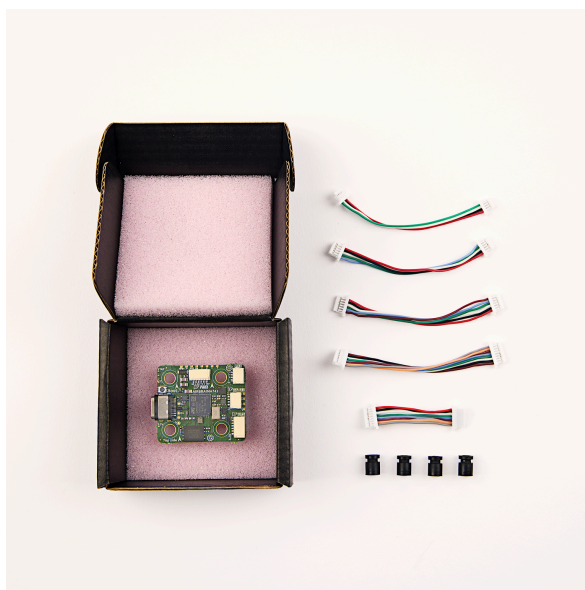
— Electrical & environmental characteristics

ELECTRICAL

Max IO voltage	3.3 V
Max VBat	42 V (10S)
Max 3.3 V current	0.5 A
Max 5 V current	2 A
Max 10 V current	2.5 A
USB cable length	≤ 2 m
Power cable length	≤ 0.5 m

ENVIRONMENT

Operating temp	0 to 60 °C
Humidity	0 to 90% RH, non-condensing
ESD handling	ESD-safe environment required
Storage temp	-20 to 70 °C



Each AirBrain H743 ships in a cardboard box with ESD-protective foam. The box contains:

Qty	Item	Used for
1 ×	AirBrain H743 flight controller	The board itself
1 ×	4-pin JST-SH cable, 60 mm soft silicone	UART2, UART4, or PX4 telemetry
1 ×	5-pin JST-SH cable, 60 mm soft silicone	PWM motor outputs
1 ×	6-pin JST-SH cable, 60 mm soft silicone	VTX, GPS, or DEBUG / SWD
1 ×	8-pin JST-SH cable, 60 mm soft silicone	ESC or AUX (long version, frame mount)
1 ×	8-pin JST-SH cable, 30 mm soft silicone	ESC short-stack version (board-on-board)
4 ×	M3 soft-silicone vibration-damping mounts	Mechanical isolation from frame

**Which firmware should I flash first?**

Use whatever your team already runs in production. The hardware definition (target AIRBRAINH743) is upstream in Betaflight, ArduPilot, and PX4, so all three flash and run out of the box.

Is it really NDAA compliant?

Yes. The bill of materials sources exclusively from EU / US / JP / TW countries; final assembly and test in Belgium.

Does it work with my existing 30×30 stack?

AirBrain H743 uses a 20×20 mount. We offer 20×20 to 30×30 adapters via our resellers; many cine FPV and mapping platforms have already retro-fitted it.

What is the warranty / RMA process?

Hardware defects: 24 months. Reach out via info@takeyourgear.com or your reseller. Field-damaged units can usually be repaired at cost.

Does it have CAN?

AirBrain H743 does not. It exposes 7 UARTs, I²C, SPI, and DShot motor outputs. A separate flight controller with CAN will join the lineup alongside AirBrain for integrators who need DroneCAN/UAVCAN. Name and timing are not announced yet, email us to be notified.



Intended use. AirBrain H743 is a flight controller for unmanned aerial vehicles. Not a toy. Install only by qualified users familiar with UAV safety. Always follow local UAV regulations.

INSTALLATION & OPERATION

- Maximum input voltage **42 V DC**. Exceeding may damage the device.
- Connect only to **3 – 10S LiPo** supported UAV power systems.
- Mount securely to minimise vibration. Do not operate near flammable materials.
- Operating envelope **0 – 60 °C, 0 – 90% RH**, non-condensing.
- Ensure correct polarity. Reverse polarity may cause permanent damage.
- Handle only in **ESD-safe environments** with proper grounding.

Regulatory & compliance

CE	EUnify BV declares conformity with EMC Directive 2014/30/EU and RoHS Directive 2011/65/EU (as amended by 2015/863). Declaration of Conformity at takeyourgear.com/pages/compliance .
EMC	Pre-compliance EMC tested at Würth Elektronik eiSos. Install and operate per this datasheet to maintain EMC performance.
RoHS	All electronic components and assemblies are RoHS-compliant.
WEEE	Subject to EU Waste Electrical and Electronic Equipment regulations. Do not dispose with household waste; return to an authorised collection facility at end-of-life.
NDAA	Supplied components are sourced exclusively from EU / US / JP / TW countries.

Disclaimer. Improper installation or misuse may result in loss of vehicle control. The manufacturer is not liable for damages resulting from misuse or modification. Always verify your build against local regulations before flight.

Sub-assembly intended for integration. This datasheet describes a sub-assembly (a flight controller / carrier board / ESC) intended for integration into a final UAV or robotics product. **CE conformity declared by EUnify BV.** Integrator declares conformity for the final product (against the EU Machinery Directive 2006/42/EC, Radio Equipment Directive 2014/53/EU, EU UAS regulation 2019/947, civil-aviation rules, or any application-specific standards). Re-test as a system if you change EMC-relevant elements (motors, ESCs, transmitters, antennas, enclosures).



DOCUMENT CONTROL

Datasheet revisions

v2.1 · 2026-05-08

Pinout colour legend; BUILD.md, LICENSE.md, PX4 board source links.

v2.0 · 2026-04-28

Full rebuild. Brand-aligned layout, expanded firmware reference, dimensions and connection diagrams.

v1.1.0 · 2025-11-05

Original datasheet.



REVISIONS

Hardware changelog

v1.0 · 2025-09

First production release. STM32H7 + clock-synced ICM-42688-P, isolated power rails, EMC-tested layout, 20 × 20 form factor, sealed USB-C.

Sales & support

Direct route to the engineering desk. Wholesale, integration, and OEM enquiries welcome.

takeyourgear.com

info@takeyourgear.com

github.com/GearUp-Company/AirBrainH743

Made in Belgium



